PROJECT REPORT

Modelling of binding of two 16-amino-acid biotinylated peptides to BSA

Peptides	A: CFAGTPSILKKNGGGS | B: KKAGTPSILMLAGGGS
Target	Bovine serum albumin (BSA), 583 aa, 66.5 kDa
Modifications	N-biotin (BTN, res 1), C-amide (NH2, res 16)
Structure prediction	Boltz-2.1 co-folding, 5 models per peptide
Binding affinity	PRODIGY (contact-based ΔG / Kd, 25 °C)
Reference project	CCB100725-YW01 (identical protocol)
Date	2026-07-14

Executive summary

Peptide A (CFAGTPSILKKNGGGS): predicted ΔG ranges -7.3 to -11.1 kcal/mol across 5 models; best model = Model 4 (ΔG -11.1 kcal/mol, Kd 6.8e-09 M, 92 interfacial contacts).

Peptide B (KKAGTPSILMLAGGGS): predicted ΔG ranges -8.1 to -9.8 kcal/mol across 5 models; best model = Model 4 (ΔG -9.8 kcal/mol, Kd 6.2e-08 M, 75 interfacial contacts).

Full per-model tables, interface-residue rankings, consensus hotspots and structure figures follow. A cross-comparison with the original peptide is given at the end.

Peptide A — CFAGTPSILKKNGGGS

Construct modeled

Biotin–CFAGTPSILKKNGGGS–NH2 (16-mer, N-biotinylated, C-amidated) bound to bovine serum albumin (BSA, 66.5 kDa, 583 aa).

Relative to the original (CFAGTPSILMLAGGGS): M10K, L11K, A12N — three central substitutions introducing two lysine residues (net +2 charge) into the C-terminal hydrophobic segment.

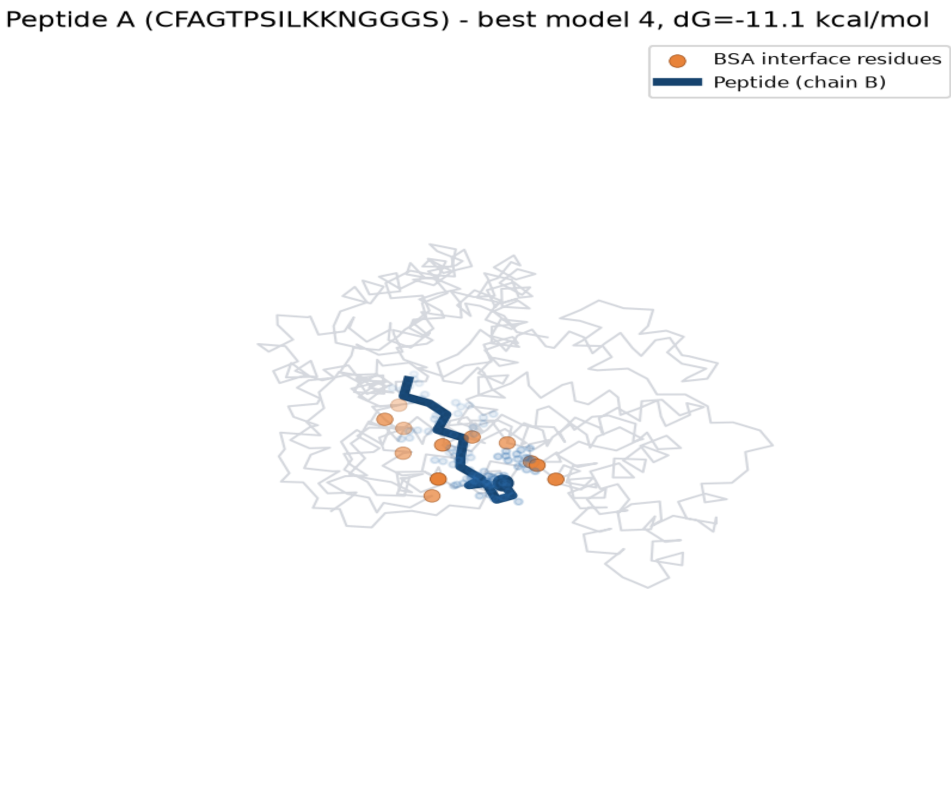
Method

Structure: Boltz-2.1 co-folding of the BSA–peptide complex (5 sampled models), biotin represented by CCD ligand BTN at residue 1 and C-terminal amidation by CCD NH2 at residue 16. Binding affinity / interface contacts: PRODIGY (contact-based predictor, 5.5 Å cutoff, 25 °C). Identical protocol to project CCB100725-YW01.

1) Model ranking and interface metrics

Model	ΔG (kcal/mol)	Kd (M @25°C)	Interfacial contacts	Hydrophobic contacts %	Apolar–apolar %
Model 1	-7.3	4.2e-06	55	70.9	21.8
Model 2	-8.6	4.9e-07	71	87.3	39.4
Model 3	-8.2	9.5e-07	56	80.4	35.7
Model 4	**-11.1**	**6.8e-09**	**92**	**84.8**	**31.5**
Model 5	-9.6	9.6e-08	73	86.3	38.4

Hydrophobic contacts % = (charged–apolar + apolar–polar + apolar–apolar)/total. Best predicted binder (lowest ΔG) highlighted.



Predicted BSA–peptide complex, best model (Model 4, ΔG -11.1 kcal/mol). BSA backbone in grey; peptide (chain B) in blue; highlighted spheres = BSA interface residues.

2) Core interface residues (top-20 per model)

Columns: BSA residue (chain A), number of contacts to the peptide, and which peptide residues it touches.

Model 1 (ΔG -7.3 kcal/mol, 55 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	THR231	4	THR5, PRO6, SER7, ILE8
2	LYS232	4	PRO6, ILE8, LEU9, LYS10
3	THR235	4	THR5, PRO6, SER7, ILE8
4	TYR262	4	LEU9, LYS10, LYS11, ASN12
5	ASN266	4	LYS10, LYS11, ASN12, GLY13
6	ASP323	4	PHE2, ALA3, GLY4, THR5
7	ARG208	3	PHE2, ALA3, GLY4
8	LYS211	3	ALA3, GLY4, THR5
9	VAL228	3	ILE8, LEU9, LYS10
10	ALA212	2	PHE2, ALA3
11	VAL215	2	ALA3, THR5
12	GLU229	2	LYS10, ASN12
13	ASP265	2	ASN12, GLY13
14	THR269	2	LYS10, ASN12
15	GLU226	1	LYS10
16	VAL230	1	THR5
17	VAL234	1	THR5
18	ALA324	1	PRO6
19	LEU326	1	PHE2
20	GLY327	1	ALA3

Model 2 (ΔG -8.6 kcal/mol, 71 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ALA324	5	PHE2, ALA3, GLY4, THR5, ILE8
2	GLU332	5	LYS11, ASN12, GLY13, GLY14, GLY15
3	GLU226	4	LYS10, LYS11, ASN12, GLY13
4	VAL228	4	PRO6, ILE8, LEU9, LYS10
5	THR231	4	GLY4, THR5, PRO6, ILE8
6	LYS211	3	PHE2, ALA3, GLY4
7	PHE227	3	GLY4, ILE8, LYS11
8	THR235	3	GLY4, THR5, PRO6
9	LEU301	3	GLY13, GLY14, GLY15
10	ASP307	3	LYS11, GLY14, GLY15
11	ASP323	3	PHE2, ALA3, GLY4
12	GLY327	3	PHE2, ALA3, GLY4
13	ARG335	3	GLY13, GLY14, GLY15
14	ARG208	2	PHE2, ALA3

15	ALA212	2	PHE2, ALA3
16	PRO302	2	GLY14, GLY15
17	SER328	2	PHE2, LYS11
18	TYR331	2	LYS11, GLY13
19	ARG336	2	GLY14, GLY15
20	VAL215	1	ALA3

Model 3 (ΔG -8.2 kcal/mol, 56 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ALA324	5	THR5, PRO6, ILE8, GLY14, GLY15
2	PHE227	4	THR5, PRO6, ILE8, GLY15
3	THR231	4	THR5, PRO6, SER7, ILE8
4	ASP323	4	PHE2, ALA3, GLY4, THR5
5	LYS211	3	PHE2, ALA3, GLY4
6	VAL228	3	SER7, ILE8, LEU9
7	ASP307	3	LYS11, ASN12, GLY13
8	PHE325	3	GLY13, GLY14, GLY15
9	SER328	3	LYS11, GLY14, GLY15
10	ARG208	2	PHE2, ALA3
11	PRO302	2	LYS11, ASN12
12	ASN317	2	GLY13, GLY14
13	ALA321	2	GLY14, GLY15
14	GLY327	2	PHE2, THR5
15	GLU332	2	LYS11, ASN12
16	ALA209	1	PHE2
17	ALA212	1	PHE2
18	VAL215	1	THR5
19	GLU226	1	LYS11
20	THR235	1	THR5

Model 4 (ΔG -11.1 kcal/mol, 92 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	GLN393	5	GLY4, THR5, PRO6, SER7, ILE8
2	LEU397	5	PRO6, SER7, ILE8, LEU9, LYS10
3	ARG409	5	PHE2, ALA3, GLY4, THR5, PRO6
4	LYS524	5	LYS11, ASN12, GLY13, GLY14, GLY15
5	GLY401	4	ILE8, LEU9, LYS10, LYS11
6	ASN404	4	ILE8, LEU9, LYS10, LYS11
7	GLU540	4	THR5, PRO6, SER7, ILE8
8	MET547	4	ILE8, LEU9, LYS10, LYS11
9	VAL551	4	LYS10, LYS11, ASN12, GLY13

10	ASP555	4	ASN12, GLY13, GLY14, GLY15
11	GLN389	3	PHE2, ALA3, GLY4
12	ASN390	3	PHE2, ALA3, THR5
13	TYR400	3	LYS10, LYS11, ASN12
14	ALA405	3	THR5, PRO6, ILE8
15	LYS520	3	GLY13, GLY14, GLY15
16	VAL554	3	GLY13, GLY14, GLY15
17	LEU386	2	PHE2, ALA3
18	LEU406	2	PHE2, THR5
19	VAL408	2	PRO6, ILE8
20	ILE512	2	GLY14, GLY15

Model 5 (ΔG -9.6 kcal/mol, 73 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU397	5	THR5, PRO6, SER7, ILE8, LYS10
2	ARG409	5	PHE2, ALA3, GLY4, THR5, PRO6
3	GLN393	4	THR5, PRO6, SER7, ILE8
4	ALA405	4	THR5, PRO6, ILE8, LEU9
5	VAL551	4	ASN12, GLY13, GLY14, GLY15
6	GLN389	3	ALA3, GLY4, THR5
7	ASN390	3	PHE2, ALA3, THR5
8	GLY401	3	ILE8, LEU9, LYS10
9	ASN404	3	ILE8, LEU9, GLY13
10	VAL408	3	PRO6, ILE8, LEU9
11	GLU540	3	PRO6, SER7, ILE8
12	LYS544	3	ILE8, LEU9, LYS11
13	LEU386	2	PHE2, ALA3
14	LYS396	2	SER7, LYS10
15	TYR400	2	GLY13, GLY14
16	LEU406	2	PHE2, THR5
17	LYS524	2	GLY14, GLY15
18	LEU543	2	ILE8, LEU9
19	ASP555	2	ASN12, GLY15
20	PHE402	1	THR5

3) BSA residues recurring across models

Seen in all/≥4 of 5 models (strong consensus): —

Seen in 3 of 5 models: ARG208, LYS211, ALA212, VAL215, GLU226, VAL228, THR231, THR235, ASP323, ALA324, LEU326, GLY327, LEU330, ALA349, GLU353

Seen in 2 of 5 models: PHE227, LYS232, VAL234, LEU301, PRO302, ASP307, LYS322, SER328, GLU332, ARG335, LEU346, LYS350, LEU386, GLN389, ASN390, GLN393, LYS396, LEU397, TYR400, GLY401, PHE402, ASN404, ALA405, LEU406, VAL408, ARG409, TYR410, ARG412, LYS413, LEU456, ARG484, PHE487, SER488,

LYS524, LEU528, GLU540, LEU543, LYS544, MET547, VAL551, VAL554, ASP555

Interface hotspot regions on BSA (residue-number bins, residues seen in ≥2 models):

- 180–209: ARG208
- 210–239: LYS211, ALA212, VAL215, GLU226, PHE227, VAL228, THR231, LYS232, VAL234, THR235
- 300–329: LEU301, PRO302, ASP307, LYS322, ASP323, ALA324, LEU326, GLY327, SER328
- 330–359: LEU330, GLU332, ARG335, LEU346, ALA349, LYS350, GLU353
- 360–389: LEU386, GLN389
- 390–419: ASN390, GLN393, LYS396, LEU397, TYR400, GLY401, PHE402, ASN404, ALA405, LEU406, VAL408, ARG409, TYR410, ARG412, LYS413
- 450–479: LEU456
- 480–509: ARG484, PHE487, SER488
- 510–539: LYS524, LEU528
- 540–569: GLU540, LEU543, LYS544, MET547, VAL551, VAL554, ASP555

4) Which peptide residues contact BSA most

Top peptide residues by interface contact count within each model.

Model	#1	#2	#3	#4	#5	#6
Model 1	PHE2 (9)	ALA3 (7)	THR5 (7)	LYS10 (7)	ASN12 (5)	PRO6 (4)
Model 2	PHE2 (14)	ALA3 (8)	GLY15 (8)	GLY4 (7)	LYS11 (7)	GLY14 (6)
Model 3	PHE2 (11)	THR5 (7)	LYS11 (7)	GLY14 (5)	GLY15 (5)	ILE8 (4)
Model 4	PHE2 (11)	ILE8 (10)	LYS11 (9)	LYS10 (8)	THR5 (7)	PRO6 (7)
Model 5	PHE2 (12)	ILE8 (11)	THR5 (8)	LEU9 (8)	PRO6 (7)	ALA3 (4)

Peptide B — KKAGTPSILMLAGGGS

Construct modeled

Biotin–(K)KAGTPSILMLAGGGS–NH2 (16-mer, N-biotinylated, C-amidated) bound to bovine serum albumin (BSA, 66.5 kDa, 583 aa).

Relative to the original (CFAGTPSILMLAGGGS): C1K, F2K — two N-terminal substitutions replacing Cys/Phe with two lysines (net +2 charge). As in the original protocol the residue-1 position is represented by the biotin cap, so the modeled construct is Biotin–K–AGTPSILMLAGGG–NH2 (first Lys represented by biotin).

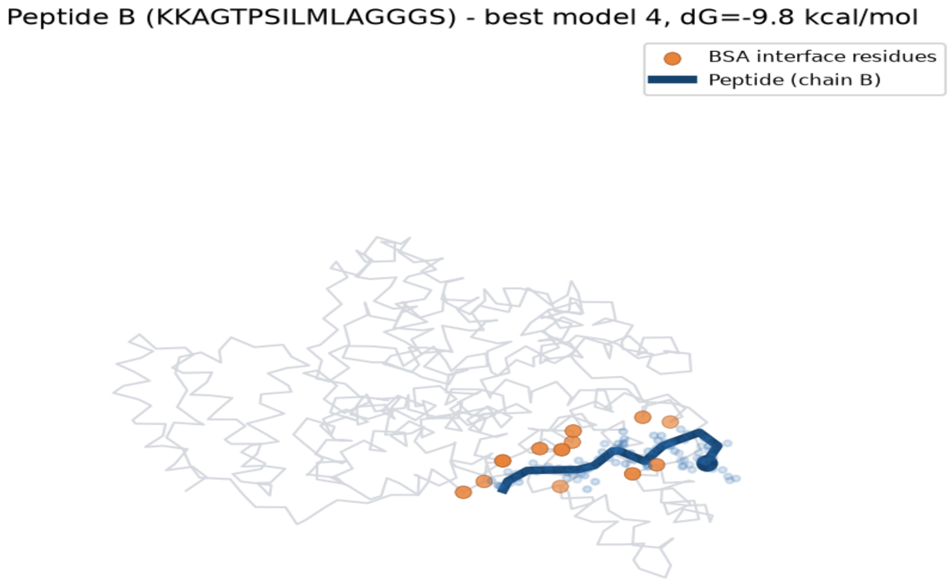
Method

Structure: Boltz-2.1 co-folding of the BSA–peptide complex (5 sampled models), biotin represented by CCD ligand BTN at residue 1 and C-terminal amidation by CCD NH2 at residue 16. Binding affinity / interface contacts: PRODIGY (contact-based predictor, 5.5 Å cutoff, 25 °C). Identical protocol to project CCB100725-YW01.

1) Model ranking and interface metrics

Model	ΔG (kcal/mol)	Kd (M @25°C)	Interfacial contacts	Hydrophobic contacts %	Apolar–apolar %
Model 1	-8.1	1.1e-06	64	90.6	39.1
Model 2	-9.2	1.9e-07	81	87.7	38.3
Model 3	-9.2	1.7e-07	80	86.2	40.0
Model 4	**-9.8**	**6.2e-08**	**75**	**92.0**	**41.3**
Model 5	-8.4	6.9e-07	78	88.5	43.6

Hydrophobic contacts % = (charged–apolar + apolar–polar + apolar–apolar)/total. Best predicted binder (lowest ΔG) highlighted.



Predicted BSA–peptide complex, best model (Model 4, ΔG -9.8 kcal/mol). BSA backbone in grey; peptide (chain B) in blue; highlighted spheres = BSA interface residues.

2) Core interface residues (top-20 per model)

Columns: BSA residue (chain A), number of contacts to the peptide, and which peptide residues it touches.

Model 1 (ΔG -8.1 kcal/mol, 64 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LYS136	5	THR5, SER7, ILE8, LEU9, MET10
2	LEU115	4	LEU9, MET10, LEU11, ALA12
3	GLU140	4	SER7, ILE8, MET10, ALA12
4	ARG144	4	ALA12, GLY13, GLY14, GLY15
5	HIS145	4	ALA12, GLY13, GLY14, GLY15
6	PRO117	3	LEU9, MET10, LEU11
7	LEU122	3	ILE8, LEU9, MET10
8	TYR160	3	LEU9, MET10, LEU11
9	ARG185	3	LEU11, ALA12, GLY13
10	LEU189	3	ALA12, GLY13, GLY14
11	GLN33	2	LYS2, ALA3
12	PRO35	2	LYS2, THR5
13	LYS116	2	LEU9, MET10
14	GLU125	2	SER7, ILE8
15	PHE133	2	ILE8, MET10
16	TYR137	2	MET10, LEU11
17	ILE141	2	LEU11, ALA12
18	GLU424	2	GLY14, GLY15
19	ARG458	2	GLY14, GLY15
20	ASP37	1	THR5

Model 2 (ΔG -9.2 kcal/mol, 81 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU115	6	SER7, ILE8, LEU9, MET10, LEU11, ALA12
2	GLU140	5	THR5, SER7, ILE8, LEU9, ALA12
3	ARG185	5	MET10, LEU11, ALA12, GLY13, GLY15
4	PRO35	4	LYS2, ALA3, GLY4, THR5
5	LYS116	4	ILE8, LEU9, MET10, LEU11
6	PRO117	4	ILE8, LEU9, MET10, LEU11
7	LEU122	4	SER7, ILE8, LEU9, MET10
8	LYS136	4	THR5, PRO6, SER7, LEU9
9	ARG144	4	ILE8, ALA12, GLY13, GLY14
10	GLU125	3	SER7, ILE8, LEU9
11	HIS145	3	ALA12, GLY13, GLY14
12	LEU189	3	LEU11, ALA12, GLY13
13	PHE36	2	GLY4, THR5
14	GLU38	2	LYS2, ALA3

15	LYS114	2	ILE8, LEU11
16	ASP118	2	ILE8, LEU9
17	TYR137	2	LEU9, MET10
18	ILE141	2	MET10, ALA12
19	TYR160	2	LEU9, MET10
20	ILE181	2	LEU9, MET10

Model 3 (ΔG -9.2 kcal/mol, 80 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ASP323	6	THR5, PRO6, SER7, ILE8, LEU9, MET10
2	ALA324	6	LYS2, ALA3, GLY4, THR5, PRO6, LEU9
3	ARG208	5	PRO6, SER7, ILE8, LEU9, MET10
4	LYS350	5	LEU11, ALA12, GLY13, GLY14, GLY15
5	LYS211	4	THR5, SER7, ILE8, LEU9
6	ALA212	4	ILE8, LEU9, MET10, LEU11
7	THR231	4	ALA3, GLY4, THR5, ILE8
8	SER328	4	LYS2, ALA3, GLY4, LEU9
9	GLU353	4	MET10, LEU11, GLY14, GLY15
10	PHE227	3	ALA3, GLY4, ILE8
11	GLU478	3	ALA12, GLY13, GLY14
12	PHE205	2	ALA12, GLY13
13	ALA209	2	MET10, ALA12
14	GLY327	2	ILE8, LEU9
15	LEU330	2	LEU9, LEU11
16	SER479	2	ALA12, GLY13
17	LEU480	2	ALA12, GLY13
18	ASN482	2	ALA12, GLY13
19	VAL215	1	ILE8
20	VAL228	1	ALA3

Model 4 (ΔG -9.8 kcal/mol, 75 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	ASP323	7	THR5, PRO6, SER7, ILE8, LEU9, MET10, LEU11
2	ARG208	6	SER7, ILE8, LEU9, MET10, LEU11, ALA12
3	PHE205	4	ALA12, GLY13, GLY14, GLY15
4	ALA212	4	LEU9, MET10, LEU11, ALA12
5	PHE227	4	ALA3, GLY4, THR5, LEU9
6	THR231	4	GLY4, THR5, PRO6, LEU9
7	ALA324	4	LYS2, THR5, PRO6, LEU9
8	ALA209	3	MET10, ALA12, GLY13
9	LYS211	3	ILE8, LEU9, MET10

10	LYS350	3	LEU11, GLY13, GLY14
11	GLU478	3	GLY13, GLY14, GLY15
12	SER479	3	GLY13, GLY14, GLY15
13	LEU480	3	ALA12, GLY13, GLY14
14	VAL481	3	ALA12, GLY13, GLY14
15	VAL228	2	ALA3, GLY4
16	THR235	2	PRO6, ILE8
17	GLU353	2	MET10, LEU11
18	THR477	2	GLY14, GLY15
19	LYS204	1	GLY15
20	VAL215	1	LEU9

Model 5 (ΔG -8.4 kcal/mol, 78 contacts)

Rank	BSA residue	Contacts	Peptide residues contacted
1	LEU115	5	LEU11, ALA12, GLY13, GLY14, GLY15
2	PRO420	5	THR5, PRO6, SER7, ILE8, LEU9
3	ARG185	4	LEU11, ALA12, GLY13, GLY14
4	LEU505	4	LYS2, ALA3, GLY4, THR5
5	LYS114	3	SER7, ILE8, LEU9
6	PRO117	3	GLY13, GLY14, GLY15
7	ILE141	3	ALA12, GLY13, GLY14
8	ARG144	3	MET10, LEU11, ALA12
9	HIS145	3	MET10, LEU11, ALA12
10	TYR160	3	GLY13, GLY14, GLY15
11	LEU189	3	MET10, LEU11, ALA12
12	LYS504	3	LYS2, ALA3, GLY4
13	ASP111	2	ILE8, MET10
14	LEU112	2	ILE8, MET10
15	LEU122	2	GLY14, GLY15
16	LYS136	2	GLY14, GLY15
17	TYR137	2	GLY13, GLY14
18	GLU140	2	GLY14, GLY15
19	ILE181	2	GLY13, GLY14
20	VAL423	2	SER7, LEU9

3) BSA residues recurring across models

Seen in all/≥4 of 5 models (strong consensus): —

Seen in 3 of 5 models: LYS114, LEU115, LYS116, PRO117, LEU122, GLU125, LYS136, TYR137, GLU140, ILE141, ARG144, HIS145, TYR160, ILE181, ARG185, LEU189, GLU424, ARG458

Seen in 2 of 5 models: PRO35, ASP37, ASP108, SER109, PRO110, ASP118, PHE133, VAL188, PHE205, ARG208, ALA209, LYS211, ALA212, VAL215, PHE227, VAL228, THR231, VAL234, THR235, ASN317, GLU320, ALA321, ASP323, ALA324, LEU326, GLY327, SER328, LEU330, LEU346, ALA349, LYS350, GLU353, GLU478, SER479,

LEU480, VAL481, ASN482

Interface hotspot regions on BSA (residue-number bins, residues seen in ≥2 models):

- 30–59: PRO35, ASP37
- 90–119: ASP108, SER109, PRO110, LYS114, LEU115, LYS116, PRO117, ASP118
- 120–149: LEU122, GLU125, PHE133, LYS136, TYR137, GLU140, ILE141, ARG144, HIS145
- 150–179: TYR160
- 180–209: ILE181, ARG185, VAL188, LEU189, PHE205, ARG208, ALA209
- 210–239: LYS211, ALA212, VAL215, PHE227, VAL228, THR231, VAL234, THR235
- 300–329: ASN317, GLU320, ALA321, ASP323, ALA324, LEU326, GLY327, SER328
- 330–359: LEU330, LEU346, ALA349, LYS350, GLU353
- 420–449: GLU424
- 450–479: ARG458, GLU478, SER479
- 480–509: LEU480, VAL481, ASN482

4) Which peptide residues contact BSA most

Top peptide residues by interface contact count within each model.

Model	#1	#2	#3	#4	#5	#6
Model 1	MET10 (9)	LEU9 (8)	LEU11 (8)	ILE8 (7)	ALA12 (7)	GLY15 (7)
Model 2	LEU9 (13)	MET10 (11)	ILE8 (10)	THR5 (7)	ALA12 (7)	LEU11 (6)
Model 3	LEU9 (11)	ILE8 (10)	ALA12 (8)	LYS2 (6)	LEU11 (6)	GLY13 (6)
Model 4	LEU9 (11)	LEU11 (9)	GLY13 (8)	GLY14 (7)	MET10 (6)	ALA12 (6)
Model 5	MET10 (10)	GLY14 (10)	GLY15 (8)	LYS2 (7)	ALA12 (7)	GLY13 (7)

5) Cross-comparison: new peptides vs. original

Peptide	Sequence	Best ΔG (kcal/mol)	Best Kd (M)	Mean ΔG	Max contacts
Peptide A	CFAGTPSILKKNGGGS	-11.1	6.8e-09	-9.0	92
Peptide B	KKAGTPSILMLAGGGS	-9.8	6.2e-08	-8.9	81
Original	CFAGTPSILMLAGGGS	-12.4	8.6e-10	-10.0	102

Predicted affinity: all three peptides are predicted to bind BSA with sub-micromolar affinity in their best models. The original peptide remained the strongest single model (ΔG -12.4 kcal/mol, Kd 8.6e-10 M). Among the new peptides, Peptide A reaches the most favourable single model (Model 4, ΔG -11.1 kcal/mol, Kd 6.8e-9 M, 92 contacts) and Peptide B is more tightly clustered and reproducible (ΔG -8.1 to -9.8 kcal/mol across all 5 models).

Binding-site behaviour — Peptide A (M10K/L11K/A12N): notably, its best-scoring Model 4 re-docks into the SAME canonical BSA pocket used by the original peptide (GLN393, LEU397, GLY401, ASN404, ARG409, GLU540, MET547, VAL551 — the 390-410 / 540-548 patch), whereas its other four models occupy an alternative groove around residues 208-353 (ARG208/LYS211/ALA212, ASP323/ALA324, GLU353). The added lysines therefore make the alternative charged groove competitive but do not abolish the original hydrophobic mode.

Binding-site behaviour — Peptide B (C1K/F2K): all five models converge on a distinct, more N-terminal/charged region of BSA around residues 114-189 (LYS114/LYS116, GLU125/GLU140, ARG144/HIS145, ARG185), recruiting Arg/Lys/Glu/His to complement the two new N-terminal lysines. This site is consistent and reproducible (higher inter-model agreement, protein-iptm up to 0.76).

Interpretation: the substitutions do not remove BSA binding; they re-route or diversify it. BSA is a promiscuous carrier with several hydrophobic and charged sub-sites, so the biotinylated peptides retain predicted sub-micromolar affinity even after inserting positive charge. All ΔG/Kd values are Boltz-2.1 (structure) + PRODIGY (affinity) predictions and should be confirmed experimentally — e.g. BLI or SPR with the biotinylated peptides captured on streptavidin sensors against BSA.

Method note: the biotin (BTN) and C-terminal amide (NH2) caps occupy peptide positions 1 and 16 respectively and are excluded from the PRODIGY contact analysis (which requires standard residues), exactly as in project CCB100725-YW01; the interface analysis therefore reflects peptide residues 2-15.